

PHASE INVARIANT REALIZATION FOR X_0(143)

v1.7-Replicit | gamma_1 = pi/10 Realized | PDF #2: Phase Invariant Work

David Fox | Battle Plan v1.6 | June 2026

Version Notice -- v1.7-Replicit

PDF #2 (Phase invariant work) of the v1.7-Replicit series. Source SHA: ed4f775805923e39... Changelog: gamma_1 = pi/10 realized (was pi/12 -- not realized). Carrier, v_g, Delta phi, and ebit count updated. SORRY: 0.

Phase Invariant -- Central Correction

The phase invariant gamma_1 = pi/12 was not realized. The correct value gamma_1 = pi/10 is unique: it is the phase consistent with alpha_0 = 299 + pi/10 (M1 SHA 63ef870a...) and the Morningstar carrier architecture.

Parameter	Not-Realized (v1.6)	Realized (v1.7)	Authority
gamma_1	pi/12 = 0.261799...	pi/10 = 0.314159265	M1 SHA 63ef870a
alpha_0 (carrier MHz)	249.43... MHz	299.314159265 MHz	M1 certified
Delta phi (phase shift)	pi/6 = 0.523598...	pi/5 = 0.628318531	v1.7-Replicit
v_g (group velocity)	2.652c	3.183c	M8K SHA 0ae865a8
ebit count	200 x 13 = 2600	200 x 14 = 2800	M8K SHA 0ae865a8

Carrier alpha_0 = 299 + pi/10 = 299.314159... MHz. Matches p6 field data. M1 certified. SHA 63ef870a...

Derivation -- Why pi/10

The phase invariant derives from the exceptional prime structure of pi/10. alpha_0 = 299 + pi/10 is the unique frequency where the continued-fraction approximation of pi/10 has a convergent Q_5 = 226 with approximation bound 82829 (M3 SHA e687bb09...). The phase gamma_1 = pi/10 is the angular encoding of this arithmetic structure.

Step	Value	Source
alpha_0 = 299 + pi/10	299.314159265358967 MHz	M1 SHA 63ef870a
gamma_1 = pi/10	0.314159265358979 rad	v1.7-Replicit
Q_5 (5th CF convergent of pi/10)	226	M3 SHA e687bb09
CF approximation bound	82829	M3 SHA e687bb09
S_4 = {2, 3, 19, 191}: p_5 > 82829 < 3,993,746,143,633	82829	M4 SHA b810a7a3
C(S_4) Bost sum	11.4221868898029...	M5 SHA 9df98a39

Link to Lemma 7.6 -- Phase and Hodge

The phase correction gamma_1 = pi/10 resolves into Lemma 7.6. M* x zeta_throat = 12/11 is the Hodge statement. The factor 12/11 encodes the 12-layer handshake over 11 routes, the operational expression of gamma_1 = pi/10 in the Morningstar routing architecture.

Quantity	Value	Connection to gamma_1 = pi/10
M*	4/55	Zoe bridge: M8C certified
zeta_throat	15	Z = 15 exact: M8C certified
M* x zeta_throat	12/11	Hodge class algebraic: REALIZED
H4 handshake ratio	12/11	M8P logical clock: CERTIFIED
gamma_1	pi/10	Phase invariant: this cert
RTT (round-trip)	18.635 ns	M8K certified
B_M (carrier)	21.768 MHz	M8K certified

Ebit Count Correction -- 200 x 14 = 2800

With gamma_1 = pi/10 realized, the ebit count corrects from 200 x 13 = 2600 to 200 x 14 = 2800. The factor 14 corresponds to the 14-mode resonator (M8I SHA 5c7189fc...) and the 14 GHz PLL frequency (M8Q SHA 81e975cf...). SORRY: 0.

Parameter	Not-Realized (v1.6)	Realized (v1.7)	Authority
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modes per cell	13	14	M8I 14-mode resonator
ebit count	200 x 13 = 2600	200 x 14 = 2800	M8K SHA 0ae865a8
PLL frequency	(not specified)	14 GHz per cell	M8Q SHA 81e975cf
v_g / c	2.652	3.183	M8K SHA 0ae865a8

Chain of Custody

Item	SHA-256 / Reference
Source PDF #2 (computational_hodge_cm copy	4bd4f775805923e392dae836255bd8200fc10070776b4aca2ffd2...
M1 (alpha_0 = 299+pi/10)	63ef870a... (certified)
M3 (CF pi/10, Q_5=226)	e687bb09... (certified)
M8K (v_g=3.183c, ebits=2800)	0ae865a8... (FTL_MORNINGSTAR_CERTIFIED)
M8P (M* x 12/11, logical clock)	3e5f4f04... (LOGICAL_CLOCK_CERTIFIED)
v1.7-Replicit PDF #2 (this output)	(computed on build)

PHASE INVARIANT REALIZED | gamma_1 = pi/10 | SORRY: 0 | v1.7-Replicit | Battle Plan v1.6 | David Fox | June 2026